

Local Model Lab Project

Document: Local Model Lab Project
Version: v1.0.0
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Date: 2026-09-06
SHA256: 7f04424be83e22fb21c785441ba809fdf122606087f404db990327895023bd8e
Chain: n/a
Tx: [not anchored]
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This is the maintained Level 3 helper. Read the [lesson](#) and [preparation](#) first. It uses the course's existing locked Python environment and Ollama, with no new account or API key.

Part	Purpose
lab.py	Fixed loopback calls, model identity checks, timed runs, private receipts, named model creation
tests/	Deterministic safety/recovery checks; no real generation
courses/project-lab/level-03/assets/prompts.json	The same writing/source/arithmetic prompts and missing-information test
courses/project-lab/level-03/assets/Modelfile.template	Starting rules copied into the learner's private folder
.zta/local-models/	Ignored private worksheet, editable Modelfile, and separate receipts for each attempt

From the course root, use `uv run --frozen zta prepare local-models, doctor, offline, benchmark, create, check, or test` in the same command position. [Command card](#) lists the complete commands.

The lab calls only `127.0.0.1:11434`, uses fixed course names, rejects cloud metadata, ignores HTTP proxy variables, and refuses redirects. It never fetches missing models, reads Level 1 keys/config, auto-selects cloud, grades answer quality, or certifies network disconnection. `create` invokes the installed Ollama CLI with an explicit loopback host and a fixed course name; rerunning it replaces that course name only.

Receipts retain model digest/bytes, full supplied prompt and hash, answer, settings, total request seconds, and incomplete/truncated status. They do not contain arbitrary learner documents. Scores, source verdicts, and observed network state require human worksheet entries. Partial attempts remain available after a failure; new attempts have distinct files. Creation does not upload or train model weights. Unload course models after use without deleting their files.

Maintainers run `uv run --frozen zta test local-models`. Live checks and honest limits belong in [Level 3 verification](#).